

**CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY**  
**1<sup>st</sup> Semester of B. Pharm. Examination**  
**University Theory Examination December 2015**  
**PH116 Pharmaceutical Biology - I**

Date: 16.12.2015, Wednesday    Time: 10:00 a.m. to 01:00 p.m.    Maximum Marks: 80

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**Instructions:**

1. There are three sections in this question paper.
  2. SECTION – I comprises of Question I. Total marks for Section I are 20. There are 20 sub-questions (MCQ type). Answers to SECTION – I are to be given in Answer Sheet for MCQ type questions provided to you. Maximum time allotted for SECTION – I is 30 minutes. Answers to SECTION – I must be written during the first 30 minutes of the examination.
  3. Answers to SECTION – II and SECTION – III are to be provided in separate Main Answer Books provided to you.
  4. Figures to right indicate marks.
  5. Draw neat sketches wherever necessary.
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**SECTION - I**

- |             |                                                      |           |
|-------------|------------------------------------------------------|-----------|
| <b>Q. 1</b> | Attempt all questions. Each question is of one mark. | <b>20</b> |
| 1.          | Gram Positive microbes stain with _____ color.       | <b>01</b> |
|             | [A] Red                                              |           |
|             | [B] Pink                                             |           |
|             | [C] Purple                                           |           |
|             | [D] Yellow                                           |           |
| 2.          | Melanocytes produce.....                             | <b>01</b> |
|             | [A] Creatinine                                       |           |
|             | [B] Bilirubin                                        |           |
|             | [C] Melanin                                          |           |
|             | [D] All                                              |           |
| 3.          | Functions of Smooth Muscle are.....                  | <b>01</b> |
|             | [A] Locomotion                                       |           |
|             | [B] Facial Expression                                |           |
|             | [C] Respiratory Movement                             |           |
|             | [D] None                                             |           |
| 4.          | Which eye structure determines a person's eye color? | <b>01</b> |
|             | [A] Cornea                                           |           |
|             | [B] Conjunctiva                                      |           |
|             | [C] Iris                                             |           |
|             | [D] Crystalline Lens                                 |           |

5. Who was the father of antiseptic surgery? 01  
[A] John Tyndall  
[B] Joseph Lister  
[C] Francesco Redi  
[D] Robert Koch
6. Who was developed the Petri dish (plate), a container used for solid culture media? 01  
[A] Francesco Redi  
[B] Robert Koch  
[C] Richard Petri  
[D] Edward Jenner
7. Removes harmful oxidants from cells is a function of \_\_\_\_\_ organelle. 01  
[A] Peroxisome  
[B] Centriole  
[C] Vacuole  
[D] Lysosome
8. Which type of photoreceptor is best able to process bright light and color? 01  
[A] Iodopsin  
[B] Rhodopsin  
[C] Cone  
[D] Rod
9. Where is the genetic material located? 01  
[A] Cell Wall  
[B] Plasmid  
[C] Nucleoid  
[D] Flagellum
10. Which organelle contains digestive enzymes? 01  
[A] Cilium  
[B] Golgi apparatus  
[C] Smooth ER  
[D] Lysosome
11. In \_\_\_\_\_ phase of Mitosis, chromosomes are pulled by the kinetochore microtubules to the poles and form a "V" shape. 01  
[A] Anaphase  
[B] Telophase  
[C] Metaphase  
[D] Prophase
12. Bulk Transport is for \_\_\_\_\_ type of molecule. 01  
[A] Glucose or Protein like large Molecule  
[B] Sodium or Potassium ion  
[C] Water  
[D] Hydrophobic Molecule

13. Which scientist was discovered first microscope? 01  
[A] Antony van Leeuwenhoek  
[B] Jan Baptista Van Helmont  
[C] Francesco Redi  
[D] John Needham
14. The site at which the axon and Muscle fiber meets known as..... 01  
[A] Mayoneuronal Junction  
[B] Trasverse Tubes  
[C] Neuromuscular Junction  
[D] A & B
15. Shape of Heart muscle is ..... 01  
[A] Long & Spindle  
[B] Long Cylindrical  
[C] Spindle Shaped  
[D] Short Cylindrical
16. I band Contain the .....protein 01  
[A] Only Actin  
[B] Only Myosin  
[C] Actin and myosin  
[D] Troponin
17. Structure which connect neighboring cells via tiny fluid filled channel known as..... 01  
[A] Connexons  
[B] Integrin  
[C] Keratin  
[D] Laminin
18. Dense regular connective tissue sorrounding entire muscle is known as 01  
[A] Epimysium  
[B] perimysium  
[C] Endomysium  
[D] None of Above
19. Function of Brown fat is..... 01  
[A] Insulation  
[B] Absorb shock  
[C] Absorb energy from sorrounding tissue  
[D] None
20. Specific Part of Sarcolemna which contains..... 01  
[A] ACH receptor  
[B] Nicotine receptor  
[C] Histamine receptor  
[D] 5 HT receptor

## SECTION – II

|             |                                                                                           |           |
|-------------|-------------------------------------------------------------------------------------------|-----------|
| <b>Q. 2</b> | <b>Attempt any <u>FOUR</u> of the following;</b>                                          | <b>20</b> |
| <b>A</b>    | Enlist a functions of following Organelles;                                               | <b>05</b> |
|             | I. Cytoskeleton                                                                           |           |
|             | II. Cilia and Flagella                                                                    |           |
|             | III. Cell wall                                                                            |           |
|             | IV. Golgi apparatus                                                                       |           |
|             | V. Lysosomes                                                                              |           |
| <b>B</b>    | Write a differences of Prokaryotic Vs Eukaryotic cell .                                   | <b>05</b> |
| <b>C</b>    | Explain Cell Division steps with suitable diagramme.                                      | <b>05</b> |
| <b>D</b>    | I. Write a short note on Passive Diffusion                                                | <b>03</b> |
|             | II. Write a short note on Osmosis                                                         | <b>02</b> |
| <b>E</b>    | Classify transport mechanisms across cell membrane. Explain simple facilitated diffusion. | <b>05</b> |
| <b>F</b>    | Classification of Microbes.                                                               | <b>05</b> |

## SECTION – III

|            |                                                                                          |           |
|------------|------------------------------------------------------------------------------------------|-----------|
| <b>Q 3</b> | <b>Attempt any <u>FOUR</u> of the following;</b>                                         | <b>20</b> |
| <b>A</b>   | Describe the following diseases;                                                         | <b>05</b> |
|            | I. Gout                                                                                  |           |
|            | II. Arthritis                                                                            |           |
|            | III. Osteoporosis                                                                        |           |
|            | IV. Fibromyalgia                                                                         |           |
|            | V. Myasthenia Gravis                                                                     |           |
| <b>B</b>   | Molecular Changes occur during muscle contraction and relaxation.                        | <b>05</b> |
| <b>C</b>   | Functions of muscular system and difference between skeletal, smooth and cardiac muscle. | <b>05</b> |
| <b>D</b>   | Describe the Sliding filament theory of muscle contraction.                              | <b>05</b> |
| <b>E</b>   | Characteristics of microbes.                                                             | <b>05</b> |
| <b>F</b>   | Staining techniques of microbes.                                                         | <b>05</b> |
| <b>Q 4</b> | <b>Attempt any <u>FOUR</u> of the following;</b>                                         | <b>20</b> |
| <b>A</b>   | Enlist loose and dense connective tissue. Explain loose connective tissue.               | <b>05</b> |
| <b>B</b>   | Classify Epithelial tissue. Explain simple epithelial tissue in detail.                  | <b>05</b> |
| <b>C</b>   | Shortnote on cell junction.                                                              | <b>05</b> |
| <b>D</b>   | Give the funtion of following cells;                                                     | <b>05</b> |
|            | I. Fibrocytes                                                                            |           |
|            | II. Fibroblast                                                                           |           |
|            | III. Macrophage                                                                          |           |
|            | IV. Microphages                                                                          |           |
|            | V. Mesenchymal cell                                                                      |           |
| <b>E</b>   | Explain in details about physiology of vision.                                           | <b>05</b> |
| <b>F</b>   | Explain in details about regulation of body temperature.                                 | <b>05</b> |